Ambient Smart Homes

Making Products for Healthy Homes

Market Opportunity

Most smart home products focus on two things: managing temperature and automating appliances, overlooking the most important function of our homes.

Providing a healthy and safe living space depends significantly on indoor air quality.

Current alternatives fall short in three main ways:

1. **Accuracy**: measure a single point at a time
2. **Convenience**: require an electrical outlet or frequent charging
3. **Insights**: do not provide a way to improve air quality

Smart Home Market:

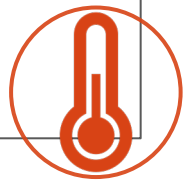
1. McKinsey: 25% of homes (~30 million) in US are 'connected'
2. Statista: current annual revenue of the Smart Home market in the US is $27 billion

Proposed Solution

Ambient plans to make a product that allows everyone to improve health of their home by monitoring and adjusting three fundamental parameters.

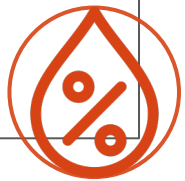
- Impacts comfort and productivity
- Correlated with humidity
- Energy intensive

Temperature



- Must be within a specific healthy range
- Correlated with temperature
- Impacts spread of contaminants

Humidity



- Organic: pollen, mold, bacteria
- Inorganic: VOCs, SVOCs, Ozone (O_3)
- Other airborne pathogens

Contaminants



3-Phase Product Roadmap

1 Monitoring

Monitors 3 key parameters through a collection of sensors

1. Monitor temperature
2. Monitor humidity
3. Monitor contaminants



Sensors + : Monitoring

2 Recommendations

Provides data-driven recommendations across 5 categories

1. Change temp
2. Change humidity
3. Increase airflow
4. Turn-on purification system
5. Change materials



Sensors + : Monitoring Recommendations

3 Automation

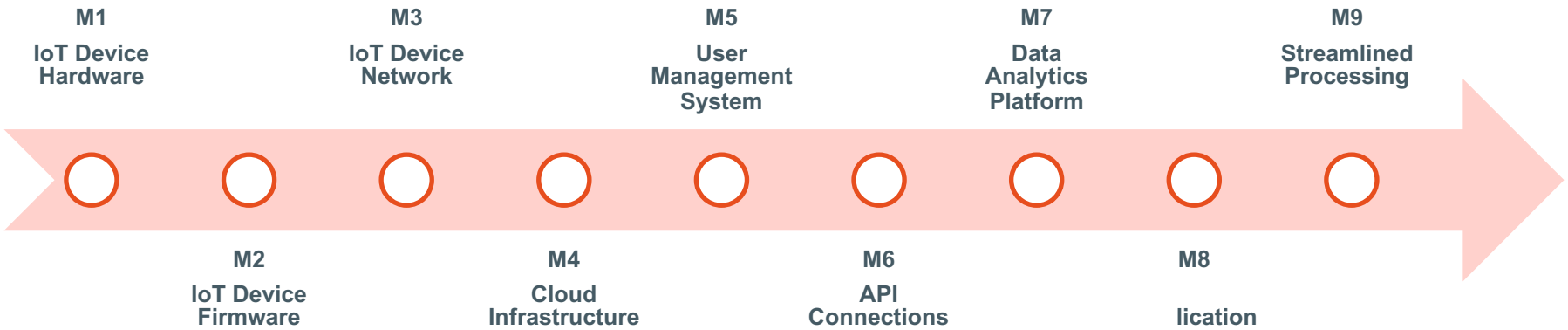
Controls 4 groups of devices to automatically improve air quality

1. Automate temperature
2. Automate humidification
3. Automate purification
4. Automate airflow



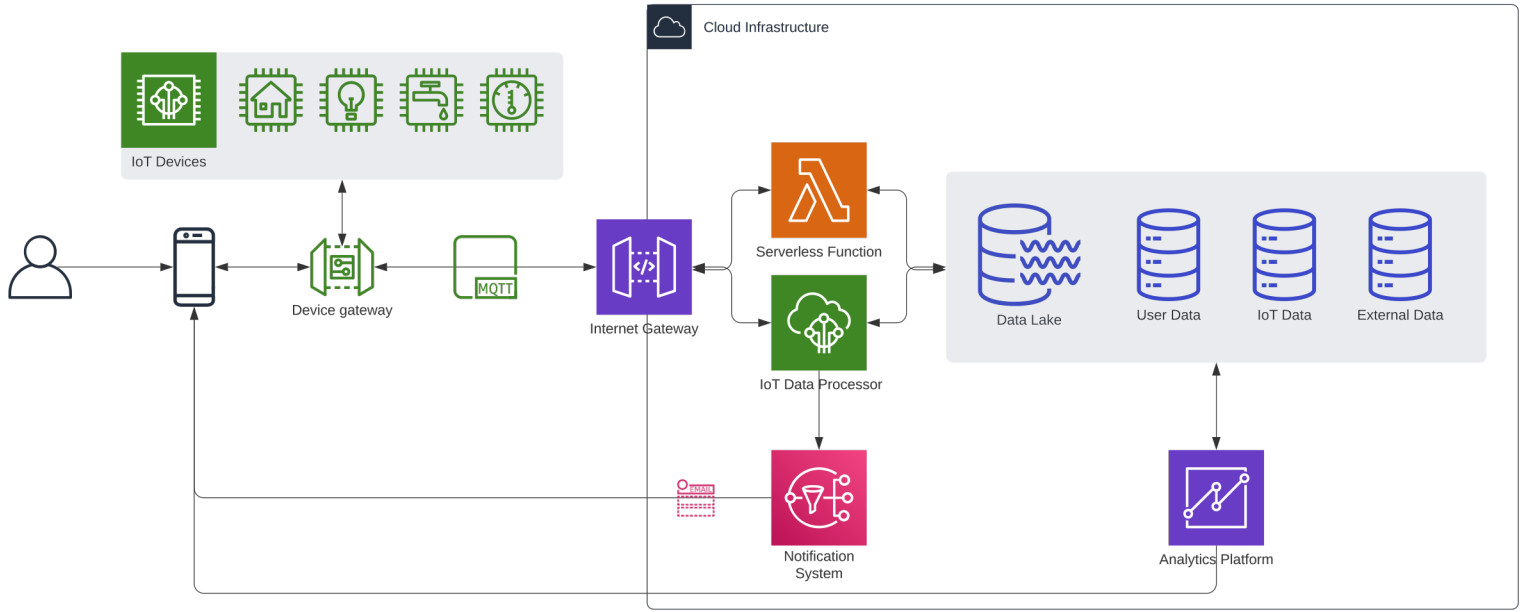
Sensors + : Monitoring Recommendations Automation + Actuators

The process for building the minimum viable product (MVP) will consist of nine key steps and take approximately one to three months, depending on hardware procurement time.



Conceptual Design

The system architecture consists of four interconnected layers: IoT devices, network connectivity, , and cloud infrastructure.

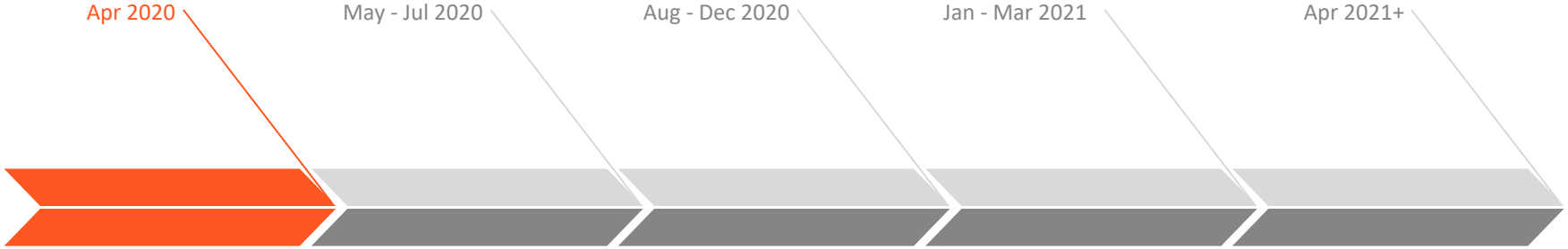


Innovation

The core innovations behind the product are the system design and technology integration, however, there are several opportunities to leverage cutting-edge research.

Layer	Component	IP Opportunity	Advantage
IoT Device	Device hardware	Highly accurate and low-cost sensors	purchase cost • Improve system accuracy
IoT Device	Device hardware	Energy efficient electronic components	operational costs • Increase convenience (charge less often)
	Device firmware	New embedded system approaches for processing data at the edge	• Real-time data processing • Lighten load on the cloud
Network Connectivity	Gateway design	A streamlined way to connect IoT devices to the cloud	hardware costs • Improve system performance
Multi-layer	lication	Analytics for accurately evaluating multiple sources of data to provide tailored recommendations through the	the need for automation • Empower users to improve air quality independently

Ambient Timeline



Planning

Establish preliminary hypothesis about the market and create a conceptual design.

Discovery

Identify the primary market, establish a network of partners and advisors, and create a prototype development plan.

GRIP Hothouse

Confirm the IP, build the prototype, and create a draft business model.

Testing

Test the Ambient platform with a select group of users and partners.

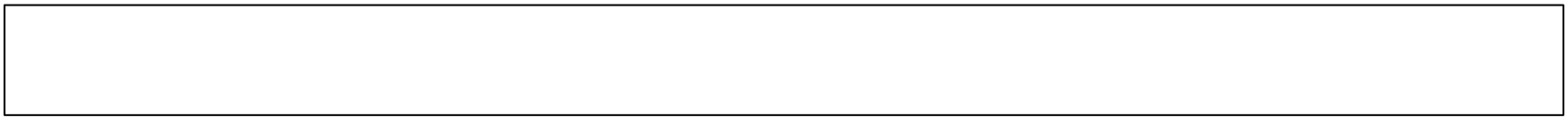
Roll-out

Confirm anchor partners, build a strong team, and roll-out first version of the product.

Thank you

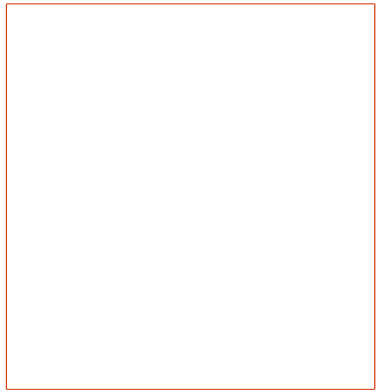
APPENDIX
(Extra Slides)

Business Model





Current Team



Ivan Damnjanovic

- 5 years in product and project management roles with tech startups
- Experienced in systems design and technology integration
- MComp, Computer Science, NUS
- BASc, Civil Engineering, U of T

IoT Device Design

Bill of Materials (BoM):

- Amazon Alexa Development Kit: USI MT7697H Development Kit for Alexa
  Processor:
  : Micro SD card 16GB
- Sensors: SDS011, HD011

Innovation

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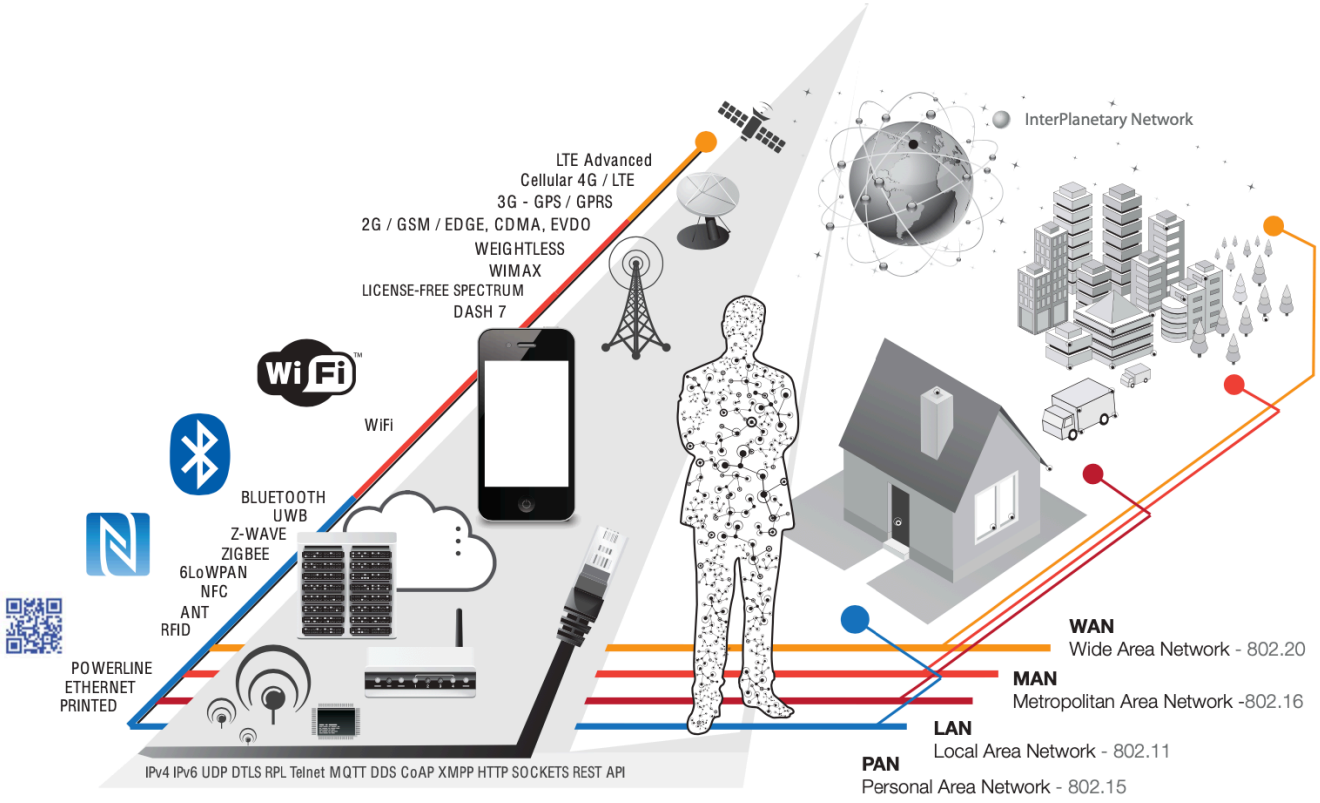
	Sensors	Energy Management	Network Connectivity	Embedded Intelligence	Analytics
Limitations	• Inaccurate • Expensive			• Processing abilities	
Potential IP				Embedded software design	
Benefits	Greater accuracy	Reduced cost			Greater customization

Competitive Analysis

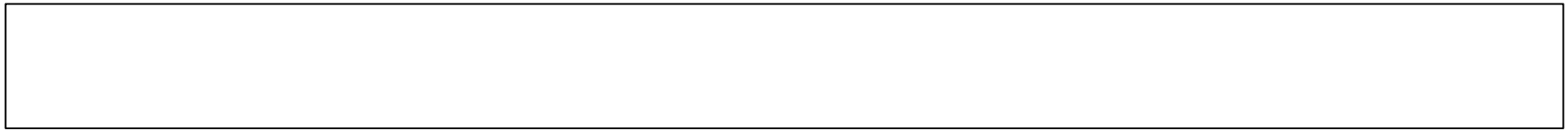
There are four categories of alternative products

Personal IAQ Products	Industrial/Commercial IAQ Products	Smart Home	Smart Buildings
Awair Plume Labs Airthings Huma-I Atmotube	Fluke Temtop Gain Express	Amazon Alexa and Echo Google Home and Nest Apple Home	Siemens LG Electronics

Technology Overview: Connectivity



User App Wireflow



Current Challenges and Limitations

There are several challenges and limitations to IoT sector.

Challenge	Mitigation
Privacy and Security	• Virtual private network • Encryption • Behavioral changes
Adoption	• Greater awareness of the connection between buildings and health • Greater comfort with new technology
Cost	• Serverless computing (compared to traditional hosting) • PaaS Services